

HOMEWORK 1

1. Prove that a finite integral domain is a field.
2. Prove that a nonzero matrix $a \in M_n(R)$, where R is an integral domain, is a zero divisor if and only if $\det(a) = 0$.
3. Let $f : R \rightarrow S$ be a surjective ring homomorphism and I a (left, right, two-sided) ideal in S . Show that $f^{-1}(I)$ is a (left, right, two-sided) ideal in R that contains $\text{Ker}(f)$. Prove that the correspondence $I \mapsto f^{-1}(I)$ yields a bijection between the set of all (left, right, two-sided) ideals in S and the set of all (left, right, two-sided) ideals in R that contain $\text{Ker}(f)$.
4. (a) An element a of a ring R is called *nilpotent*, if $a^n = 0$ for some $n \in \mathbb{N}$. Show that if R is a commutative ring, then the set $\text{Nil}(R)$ of all nilpotent elements in R is an ideal (*the nilradical of R*). Prove that the quotient ring $R/\text{Nil}(R)$ has no nonzero nilpotent elements.
(b) Prove that a polynomial $f(t) = a_0 + a_1t + \dots + a_nt^n \in R[t]$, where R is a commutative ring, is nilpotent if and only if all a_i are nilpotent in R .
5. (a) Prove that if a is a nilpotent element of a ring R , then the element $1 + a$ is invertible.
(b) Prove that a polynomial $f(t) = a_0 + a_1t + \dots + a_nt^n \in R[t]$, where R is a commutative ring, is invertible in $R[t]$ if and only if a_0 is invertible in R and a_i are nilpotent in R for $i \geq 1$.
6. Let R be a ring and $r \in R$. Prove that there is a unique ring homomorphism $f : \mathbb{Z}[t] \rightarrow R$ such that $f(t) = r$. Show that the image of f is the smallest subring of R that contains r . Prove that the forgetful functor $\text{Rings} \rightarrow \text{Sets}$ is representable.
7. Let R be an integral domain such that $R[t]$ is a principal ideal domain. Prove that R is a field.
8. Prove that for every nonzero commutative ring R , the ring $R[t]$ has infinitely many prime ideals.
9. The *Jacobson radical* of a commutative ring R is the ideal $\text{Rad}(R)$ that is the intersection of all maximal ideals in R . Show that $x \in \text{Rad}(R)$ if and only if $1 - xy \in R^\times$ for all $y \in R$.
10. Determine the left adjoint functor to the forgetful functor $\text{CommRings} \rightarrow \text{Sets}$.